

FINAL EXPLANATION: BIOLOGY GRADE 11

Student Assessment Document

FINAL EXPLANATION: BIOLOGY GRADE 11: GROWTH AND DEVELOPMENT IN PLANTS (SUB-STRAND 2.2)

Student Assessment Document

Student Name

Class

Date

INSTRUCTIONS FOR STUDENTS

You are a scientist helping a farmer in Nakuru County. The farmer planted maize and beans on the same day after the long rains. Some seeds germinated within a few days, some took longer, and some healthy-looking seeds never sprouted. As the plants grew, they also differed in height, thickness, root depth and branching. Your task is to write a Claim-Evidence-Reasoning (CER) explanation that answers the driving question: What causes the differences in germination, emergence and growth of maize and bean plants? Use evidence from our unit: your seed drawings and dissections, the dormancy and germination investigations, the stem sections, the growth curves and the hormone notes. Write your answers in the space under each prompt. Use labelled drawings and arrows where they help. In Section 4 you will also revise your Day 1 model. Write in full sentences and use the scientific words we have learned.

Section 1: Seed structure and food supply

Compare a maize grain and a bean seed. (a) Name the main structures you saw in each, and say what each one does. (b) Describe how these two seeds differ in the number of cotyledons and in where the food is stored. (c) Explain how these differences could affect the way each seedling emerges and grows early on. Refer to your dissections of soaked seeds.

Write your answer here:

Section 2: Dormancy and germination conditions

Some of the farmer's seeds looked healthy but did not germinate, and others were late. Write a CER explanation using the two investigations we did. **Claim:** give two different reasons why seeds planted after the same rain may germinate late or fail. **Evidence:** describe what you observed in the dormancy treatments (control, sandpaper, warm water, leaching or chilling) and in the germination sets (water, air and warmth; dry; little oxygen; cool; boiled seeds). **Reasoning:** explain why each result happened, using the words dormancy, viable, water, oxygen and temperature.

Write your answer here:

Section 3: How the seedlings emerge and keep growing

(a) Compare how maize and bean seedlings emerge from the soil. Use the terms hypogeal and epigeal. (b) Explain how primary growth and secondary growth differ, and where in the plant each one happens. (c) Use your stem-section observations to explain why a bean plant can become thicker and branch while maize grows mainly in length. (d) Describe the shape of the growth curve you plotted and what it tells you about growth over time.

Write your answer here:

Section 4: Hormones and your final CER explanation

Write your final explanation to answer the driving question: What causes the differences in germination, emergence and growth of maize and bean plants?
Claim: state your answer in one or two sentences. **Evidence:** give at least three pieces of evidence from different lessons, including at least one about hormones (for example, the bean seedling with its tip removed). **Reasoning:** link your evidence to your claim and explain how hormones work together with water, oxygen, temperature and seed structure. Finally, say how your Day 1 model or drawing has changed, and describe one thing you added.

Write your answer here:

Section 5: Applying your understanding

A farmer in Kakamega or Nyeri wants to improve germination and rooting for a school tree nursery. Give two practical pieces of advice, and explain each one using ideas from this unit. Mention one natural rooting source you tested.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Scientific accuracy	All ideas are correct: seed structure, dormancy, germination conditions, primary and secondary growth, and hormone roles, with correct use of terms such as monocot, dicot, hypogeal, epigeal and meristem.	Most ideas are correct, with one or two small errors or terms used loosely.	Several ideas are incorrect or confused, for example mixing up primary and secondary growth or seed types.
Use of evidence	Gives at least three specific pieces of evidence from different lessons (seed dissections, germination and dormancy sets, stem sections, growth curves, tip-removal bean seedling).	Gives two relevant pieces of evidence, or three that are general.	Gives little or no evidence, or evidence that is not from our investigations.
Reasoning	Clearly explains how each piece of evidence supports the claim and links seed structure, conditions, dormancy, meristems and hormones together.	Explains some links between evidence and claim, but some factors are listed rather than explained.	Mostly lists facts without explaining how they answer the question.
Connection to the driving question and revised model	Directly answers why maize and beans planted together differ in germination, emergence and growth, and clearly shows how the Day 1 model was revised	Answers the question and mentions a model change with few details.	Does not clearly answer the question or does not describe any change to the model.

	with specific additions.		
Clarity of communication	Well organised in claim, evidence and reasoning, with full sentences, clear labelled drawings and correct vocabulary.	Mostly clear, with some gaps in organisation or labelling.	Hard to follow, with missing structure or unlabelled drawings.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

Licensing & Copyright: © 2026 SeaVuria and ARES. This work is licensed under a [Creative Commons Attribution-NonCommercial 4.0 International License \(CC BY-NC 4.0\)](https://creativecommons.org/licenses/by-nc/4.0/). To view a copy of this license, visit <https://creativecommons.org/licenses/by-nc/4.0/>. You are free to copy, redistribute, and adapt this material in any medium or format, provided appropriate credit is given and the material is not used for commercial purposes. KICD learning outcomes and curriculum designations reproduced here for alignment purposes remain the property of KICD and are not covered by this license. Linked external resources (e.g. CK-12, Khan Academy, PhET) are subject to their own licenses.
